
SAILBOAT NAVIGATION AND DOCKING ASSIST

TECHNICAL AREAS

Sensors, Control Systems PCB Design, Solar Power, Power Management System, Data Visualization, iOS, BLE, 3D Printing/Mechanical Design, Sailing Physics

CUSTOMER'S PROJECT DESCRIPTION

Sailing in a crowded harbor or racing alongside other boats in close proximity is challenging, especially when sailing single handedly. Oftentimes, the skipper's field of view is blocked by the main and front sails, so it is difficult to maintain a visual on obstructions or other boats. Additionally, docking under sail or even under power can present difficulties to the skipper, under many conditions, such as unfavorable wind direction, current, or limited visibility. This project aims to develop a robust and portable sensor system that will aid sailors while underway and docking, and present the status to the user via a BLE connected iOS app. The sensor system can potentially be used in the future to optimize skipper sailing performance.

HIGH-LEVEL REQUIREMENTS

- 1) The sensor system shall report proximity to nearby objects and boats via an audio and visual alert.
- 2) The sensor system shall report speed over ground, speed through water, wind speed/direction.
- 3) The system shall include real-time collision avoidance alerts, and as a bonus, follow right-of-way (COLREGs) rules.
- 4) The sensor system shall be easily installed/removed from all types of boats without causing damage.
- 5) The sensor system shall have reliable BLE communication with an iPhone.
- 6) The system shall contain multiple sensor types:
 - a. Distance sensors (with Object Detection)
 - b. GPS, compass, speed-over-ground
 - c. Wind speed (Anemometer)
- 7) The system shall be powered via a rechargeable battery, as most small sailboats don't have accessible power.
- 8) The status shall be reported in a visually pleasing iOS app.
- 9) The app shall report to the user recommended changes in course based on sensor readings.
- 10) The system shall be designed to be dust/waterproof rated to IP68 (by design, but bonus points for test report) and able to withstand harsh marine environments.

DELIVERABLES

The deliverable shall be:

1. A sensor hardware system as defined above, with custom housing
2. An iOS application
3. System software, version-controlled in GitHub
4. Test results from an actual field test on a sailboat

CUSTOMER'S CONTACT INFORMATION

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